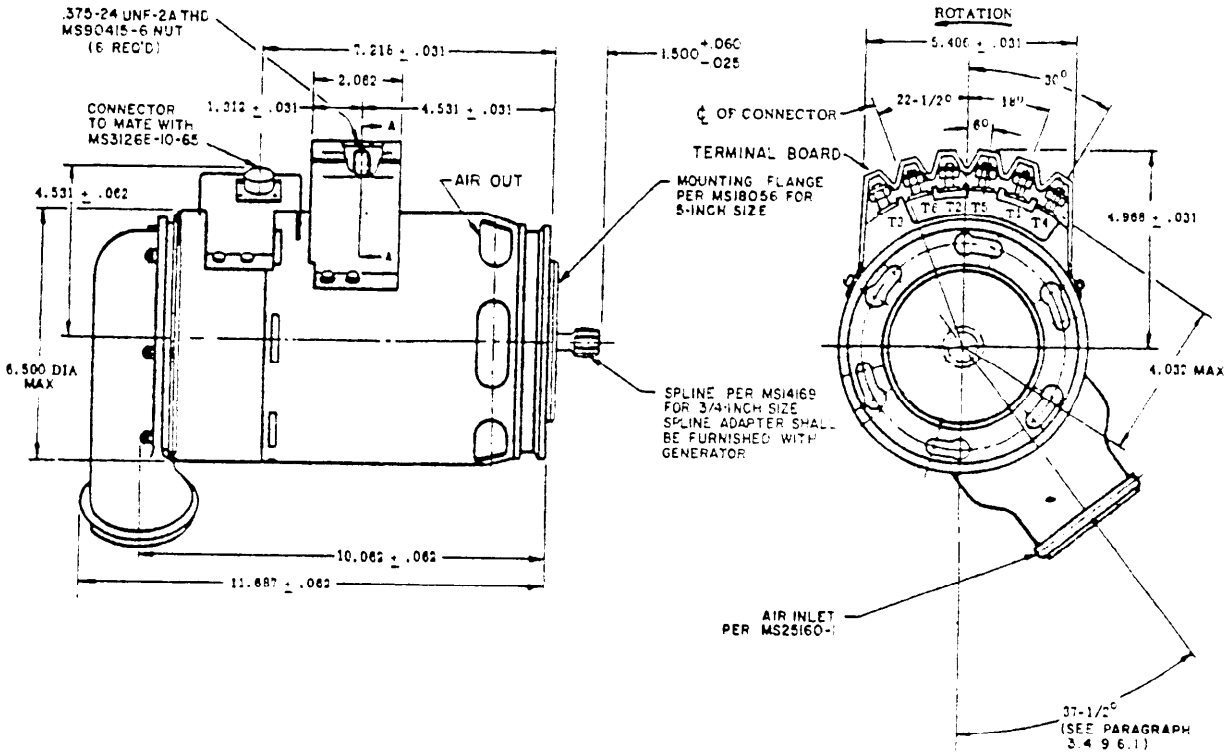
MILITARY SPECIFICATION SHEET

GENERATOR SYSTEM, 10 K.V.A., 400 HERTZ, ALTERNATING CURRENT, INTEGRALLY EXCITED BRUSHLESS TYPE

This specification is approved by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the generator described herein shall consist of this document and the issue in effect of specification MIL-G-21480.



GENERATOR DATA	
RATED VOLTAGE	120/208V
RATED OUTPUT	10 KVA
PHASE	3
FREQUENCY	380-420 CPS
SPEED RANGE	7600-8400 RPM
MAX. SPEED FOR REGULATION	10,000 RPM
OVERSPEED	11,000 RPM
RATED POWER FACTOR	.75/lag. to 1.0
EFFICIENCY, MIN. AT RATED LOAD	80%
MAX. WT.	30 LBS
OVERHUNG MOMENT - MAX	125 LB. IN.
SHEAR	650-750 IN. LB
FLEXIBLE DRIVE	NOT REQUIRED

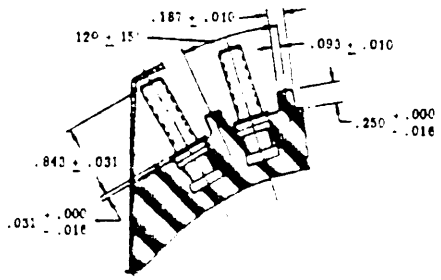


FIGURE 1. M21480/7-1, GENERATOR

FSC 6115

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

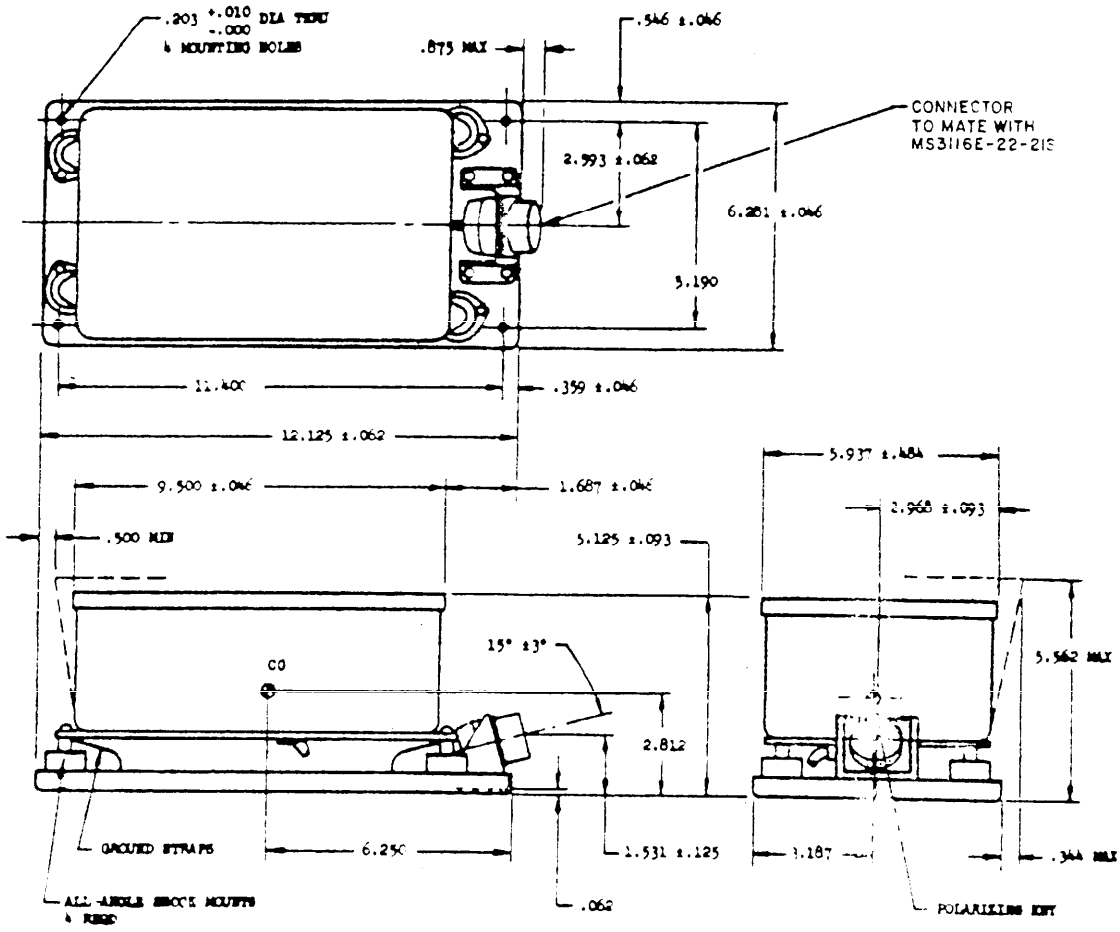


FIGURE 2. M21480/7-3, GENERATOR SUPERVISORY UNIT, WT [illegible] LBS MAX

Source: https://assist.dla.mil -- Downloaded: 20260-19T09:41Z
Check the source to verify that this is the current version before use.

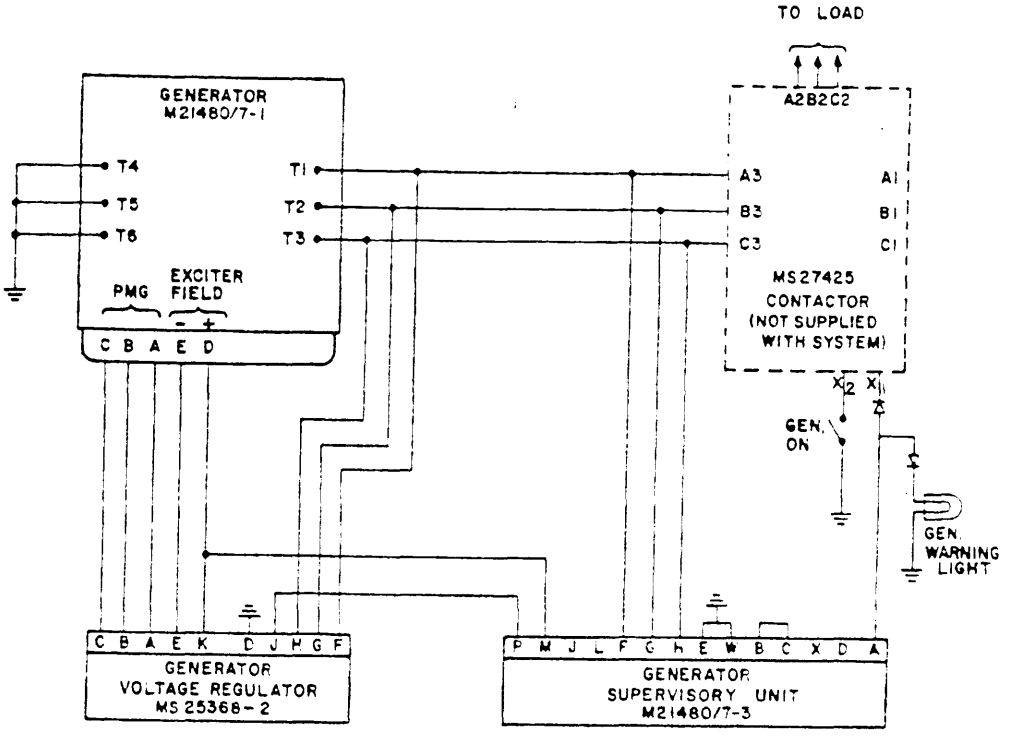


FIGURE 3. TEST CIRCUIT

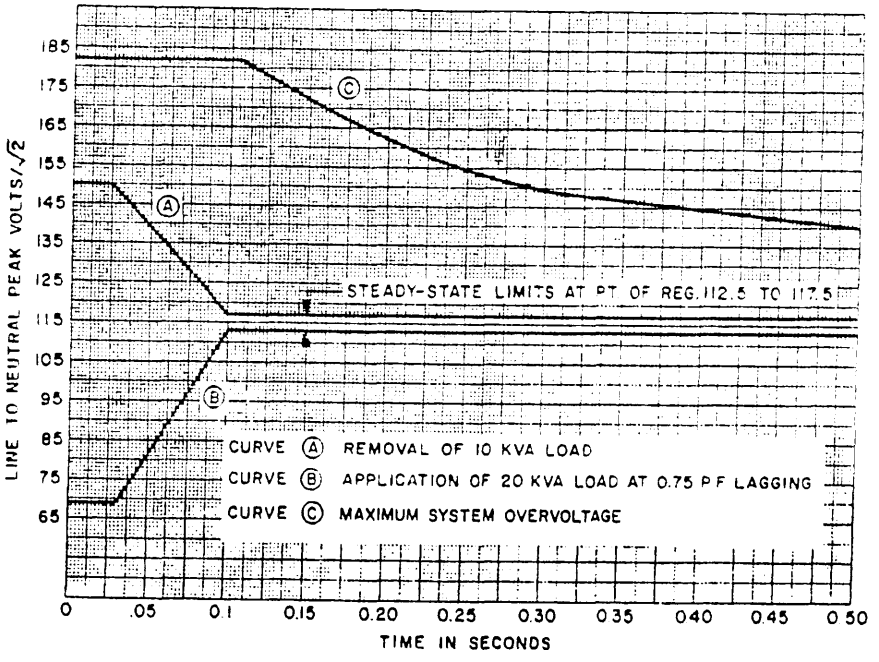


FIGURE 4. TRANSIENT, STEADY-STATE AND OVERVOLTAGE LIMITS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

The requirements of MIL-G-21480 apply with the following exceptions:

Delete the following paragraphs:

3.4.2.2 through 3.4.2.8, 3.4.6, 3.4.10, 3.4.10.5, 3.4.10.6, 3.4.13.1, 3.4.15, 3.4.16, 3.4.16.1 through 3.4.16.3, 3.5.7.5, 3.5.7.6, 4.5.20, 4.5.22.1 through 4.5.22.3, 4.5.22.3.1 through 4.5.22.3.4, 4.5.24.1 through 4.5.24.3, 4.5.24.3.1 through 4.5.24.3.4, 4.5.25.1, 4.5.25.2, 4.5.25.2.1, 4.5.28, 4.5.32, 4.5.33, 4.5.33.1, 4.5.33.2.

3.4.2 Delete and add: Environmental requirements. The system shall meet the requirements of MIL-E-81910 unless otherwise specified by the detailed specification.

3.4.2.1 Delete and add: Temperature and altitude. The generator shall fully support the system when subjected to the temperature-altitude requirements of MIL-E-81910 for blast cooled equipment. The control components shall operate continuously when subjected to the temperature-altitude requirements of MIL-E-81910 for self-cooled equipment.

3.4.8 Change "Figure 3" to "Figure 4 of the detailed specification".

3.4.8.2 Change "5 minutes" to "2 minutes".

3.4.9.3 Delete and add: Electric connections. The generator shall be supplied with terminals as shown in the detailed specification. A protective cover shall be provided to cover the main terminal block and all attached terminal lugs. The cover shall withstand normal handling and installation without damage.

3.4.9.6.1 Delete last sentence and add: The air inlet duct shall be positionable in increments of 60° from the position shown in Figure 1 of the detailed specification.

3.4.9.6.1.1 Change "10,000 hour life" to "1000 hour life".

3.4.9.7 Change "10,000 hours" to "5000 hours".

3.4.17 Delete and add: Reliability. The system MTBF (Mean Time Between Failures) shall be not less than 1200 hours. This figure shall apply to failures which will cause system operation to deviate from performance requirement.

3.5.1 Change "Figure 3" to "Figure 4 of the detailed specification".

3.5.4 Delete and add: Wave form. The crest factor for each phase voltage wave form shall be 1.41 ± 10%. No single harmonic shall exceed 3% of the fundamental.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

3.5.5 Delete "5 amps minimum continuous capacity".

3.5.7.2 Change "Figure 6" to "Figure 4 of the detailed specification".

3.5.7.3 Delete and add: Undervoltage. Undervoltage protection shall be provided by a static element which functions to disconne the system from the load bus and deenergize the generator within a period of 4 ± 1 seconds after any phase voltage (line-neutral) falls to 90-105 volts. The system shall be capable of being reset within 0.5 second when each phase voltage is above dropout and less than 105 volts.

3.5.7.4 Delete and add: Underfrequency. Underfrequency protection shall be provided by a static element which functions to disconnect the system from the load bus within a period of 4 ± 1 seconds after the frequency falls to 377-380 Hz. Pick up shall occur within 0.5 second at 380-383 Hz.

3.5.8 Delete and add: Electromagnetic interference. The system shall meet the requirements of MIL-E-81910 Class No. IIIB for the generators and control equipment, except the ten microfarad feed-through capacitor shall be removed.

4.2.1.1 Add: Qualification tests shall be supplemented with a minimum 100 hour failure-free flight test before consideration is given to incorporating the item on the Qualified Products List.

4.5.3.1 Change "Figure 3" to "Figure 4 of the detailed specification".

4.5.3.2 Change "Figure 3" to "Figure 4 of the detailed specification".

4.5.12 Delete and add: Electromagnetic interference. The system shall be subjected to an electromagnetic interference test in accordance with MIL-E-81910 for Class IIIB equipment.

4.5.13 Delete and add: Wave form. The crest factor and harmonic content, line-line and line-neutral, of the output voltage shall be determined under the following conditions:

a. At generator speed of 8000 RPM for no load, and in 20% load increments to full rated load, unity power factor.

b. Condition (a) shall be repeated for 0.75 power factor.

c. Conditions (a) and (b) shall be repeated at generator speeds of 7600 and 8400 RPM.

The following requirements must be met during all above tests. The line-neutral crest factor shall be 1.41 ± 10%. No harmonic shall exceed 3% of the fundamental.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

4.5.20.1 Delete and add: Salt fog, control components. Control components shall be subjected to a Salt Fog Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal or rated and 200% load, and protective functions.

4.5.20.2 Delete and add: Oil-salt water ingestion, generators. The generator shall be subjected to an Oil-Salt Water Ingestion Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

4.5.21 Delete and add: Fungus resistance. System components shall be subjected to a Fungus Test in accordance with MIL-E-81910, with the installation instructions and mounting nuts attached. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions. Mounting nuts are to be unaffected and the package holding the nuts suitable for shipment. Instructions are to be legible.

4.5.22 Delete and add: Sand and dust. The generator shall be subjected to a Sand and Dust Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

4.5.23.1.1 Delete and add: Generator. The generator shall be subjected to a Vibration Test in accordance with MIL-E-81910, except that the amplitude of vibration monitored at the anti-drive end of the generator during the main bending mode shall be limited to 25g's but the vibration input shall not be lowered below 5g's. At the completion of this test the unit shall successfully demonstrate its buildup characteristics and the effect of application and removal of rated and 200% load.

4.5.23.1.2 Delete and add: Control components. Control components shall be subjected to a Vibration Test in accordance with MIL-E-81910, except that the vibration range shall be from 5 to 500 Hz. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.24 Delete and add: Shock. System components shall be subjected to a Shock Test in accordance with MIL-E-81910. At the completion of procedure I, the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions. There shall be no failure of the mounting attachments and the test item shall remain in place and not create a hazard during procedure III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

4.5.25 Delete and add: Humidity. System components, except those tested under paragraph 4.5.20.2, shall be subjected to a Humidity Test in accordance with MIL-E-81910. At the completion of this test the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

4.5.30 Delete first sentence and add: All system components except the generator shall be subjected to the Acceleration Tests of MIL-E-81910. At the completion of the tests the unit shall successfully demonstrate its buildup characteristics, the effect of application and removal of rated and 200% load, and protective functions.

NOTES:

1. Dimensions in inches. Unless otherwise specified, tolerances: decimals ± .005, angles ±1°.
2. Protrusions such as rivets and screw heads shall be contained within the outline shown.
3. Where only maximum dimensions are shown, the part need not have the shape shown but must be contained within the outline shown.

ITEM	PART NUMBER
Complete System (Consists of one Generator, one Generator Voltage Regulator, and one Generator Supervisory Unit)	M21480/7-1
Generator	M21480/7-10
Generator Voltage Regulator	MS25368-2
Generator Supervisory Unit	M21480/7-3

For design feature purposes, this standard takes procedence over procurement documents referenced herein. Referenced documents shall be of the issue in effect on date of invitations for bid.

Preparing Activity:
Navy - AS
Project No. 6115-N387

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use submitting comments and suggestions for improvements. All users of military standardization documents are invited to provi suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, ar mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, w too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAG
NECESSAR
IF MAILED
IN THE
UNITED STAT

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering Standardization Department
(SESD). Code 93
Lakehurst, NJ 08733

ATTN: TJM (9322)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:41Z
Check the source to verify that this is the current version before use.